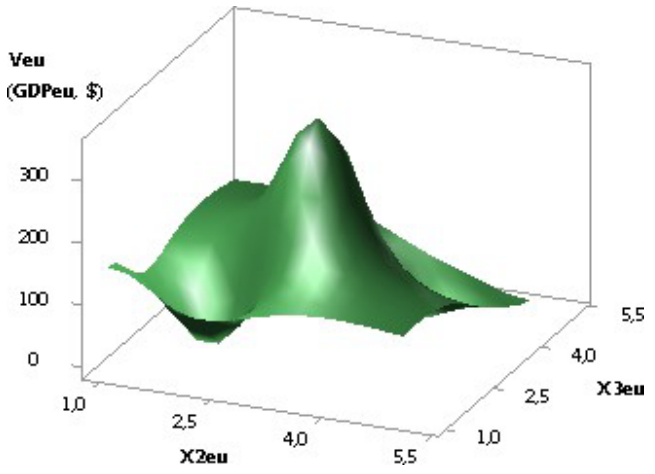
Э. А. ПИЛЬ

ТЕОРИЯ ЭКОНОМИЧЕСКИХ КРИЗИСОВ

VI

THE THEORY OF ECONOMIC CRISES

VI



САНКТ-ПЕТЕРБУРГ
2024

CONTENTS

INTRODUCTION 5

CHAPTER 1
APPLICATION OF AUTOMATIC CONTROL THEORY FOR A COUNTRY'S EXIT FROM AN ECONOMIC CRISIS 6

The Theory of a country's recovery from an economic crisis 6

1.1. The Theory of Automatic Control 6

Key concepts 6

1.2. Classification of Automatic Control Systems 7

1.3. Classification of adaptive (Self-Tuning) systems 9

The Concept of adaptive control systems (variable structure systems) 10

Principles of ACS control 11

Mathematical models of linear ACS 11

Transfer function of a linkage of elements 12

Linearization of systems and components 13

Controllability and Observability of ACS 14

Stability of linear systems 14

Stability criteria 15

Stability margin of ACS 17

Criteria 17

Comparative analysis of stability criteria 17

Dynamic link 18

Typical dynamic elements 18

Regulation static and dynamics 19

Relationships between dynamic characteristics 20

1.4. Automatic Control System for economic crisis 21

Measures taken in Russia during the 2008 economic crisis 22

1.5. Types of stabilization funds and the necessity of their formation 22

Eurasian fund for stabilization and development 23

Analytical activities 23

Stabilization fund of the Russian Federation 25

National wealth fund of Russia 24

What is the national wealth fund of Russia? 24

Main objectives of the creation of the national wealth fund 24

Covering the federal budget deficit 24

In 2023, the liquid part of reserves decreased by one-third 25

CONCLUSIONS 25

CHAPTER 2
CALCULATION OF THE UPPER CRITICAL VOLUME *Vsu* OF A SPHERICAL ECONOMIC SHELL AT NEGATIVE VALUES OF VARIABLES 26

1.1. Calculation of the upper critical volume *Vsu* of a spherical economic shell with negative values of variables 26

1.2. Summary tables of the upper critical volume *Vsu* of a spherical economic shell with negative values of variables 112

CONCLUSIONS 123

CHAPTER 3
CALCULATION OF THE LOWER CRITICAL VOLUME *Vsl* OF A SPHERICAL ECONOMIC SHELL AT NEGATIVE VALUES OF VARIABLES 124

1.1. Calculation of the lower critical volume *Vsl* of a spherical economic shell with negative values of variables 124

1.2. Summary tables of the lower critical volume *Vsl* of a spherical economic shell with negative values of variables 157
CONCLUSIONS 167

CHAPTER 4
CALCULATION OF THE UPPER CRITICAL VOLUME OF *Veu* AND THE VARIABLE X5*eu* OF THE ELLIPSOID ECONOMIC SHELL AT NEGATIVE VALUES OF OTHER VARIABLES 168
1.1. Calculation of the upper critical volume *Veu* and the variable X5*eu* of an ellipsoidal economic shell with negative values of other variables 168
1.2. Summary tables of the upper critical volume *Veu* and the variable X5*eu* of an ellipsoidal economic shell with negative values of other variables 253
CONCLUSIONS 261

CHAPTER 5
CALCULATION OF THE UPPER CRITICAL VOLUME OF *Veu* AND THE VARIABLE X6*eu* OF THE ELLIPSOID ECONOMIC SHELL AT NEGATIVE VALUES OF OTHER VARIABLES 262
1.1. Calculation of the upper critical volume *Veu* and the variable X6*eu* of an ellipsoidal economic shell with negative values of other variables 262
1.2. Summary tables of the upper critical volume *Veu* and the variable X6*eu* of an ellipsoidal economic shell with negative values of other variables 425
CONCLUSIONS 435

CHAPTER 6
CALCULATION OF THE LOWER CRITICAL VOLUME OF *Vel* AND THE VARIABLE X4*el* OF THE ELLIPSOID ECONOMIC SHELL AT NEGATIVE VALUES OF OTHER VARIABLES 436
1.1. Calculation of the lower critical volume *Vel* and the variable X4*el* of an ellipsoidal economic shell with negative values of other variables 436
1.2. Summary tables of the lower critical volume *Vel* and the variable X4*el* of an ellipsoidal economic shell with negative values of other variables 493
CONCLUSIONS 500

CHAPTER 7
CALCULATION OF THE LOWER CRITICAL VOLUME OF *Vel* AND THE VARIABLE X5*el* OF THE ELLIPSOID ECONOMIC SHELL AT NEGATIVE VALUES OF OTHER VARIABLES 501
1.1. Calculation of the lower critical volume *Vel* and the variable X5*el* of an ellipsoidal economic shell with negative values of other variables 501
1.2. Summary tables of the lower critical volume *Vel* and the variable X5*el* of an ellipsoidal economic shell with negative values of other variables 558
CONCLUSIONS 565

LIST OF ABBREVIATIONS 566

REFERENCES 567

BIOGRAPHY 580

'For me a word of wisdom is more valuable than any gold.'
Democritus

INTRODUCTION

Various authors dedicated a huge amount of their works to the issue of economic and financial crisis and offered their ways out of the crisis. Although, unfortunately, none of works succeeded in the profound solving of the problem. Due to this fact the author of this work suggests to use the theory of the economic shell in order to describe processes in economy suffering from crisis. This theory gave an opportunity to design theoretic scenarios of economic crisis development and to perform calculations which help to indicate the probable methods to be used in recovering from any type of economic recession. These methods enable government to choose an appropriate way out of the particular crisis without loss and even with the increase of gross domestic product of a country.

The book is dedicated to my son *Mikhail* and grandson *Stepan*, and to my late beloved parents and elder brother.

Wishing you all the best.

Yours sincerely, Author



[172]